MILITARY SPECIFICATION

RUST ARRESTING COATING (FOR TREATMENT OF RUSTED METAL)

This specification is approved for use by the U.S. Army Materials Technology Laboratory, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a clear (unpigmented) coating based on vegetable and animal oils for use on rusted metal and painted surfaces to arrest further corrosion. It provides two compositions, one of which is suitable for use under AIR POLLUTION REGULATIONS (see 6.5).

1.2 Classification. The rust arresting coating covered by this specification shall be of the following composition as specified:

Composition G - General use.
Composition L - Limited use (see 6.5).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, US Army Laboratory Command, Materials Technology Laboratory, ATTN: SLCMT-MSE, Watertown, MA 02172-0001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A /FSC 8030/

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SPECIFICATIONS

FEDERAL

PPP-P-1892 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing and Marking of
TT-T-291 - Thinner; Paint, Volatile Spirits, Petroleum Spirits

STANDARDS

FEDERAL

FED-STD-141 - Paint, Varnish, Lacquer, and Related Materials, Methods of Inspection, Sampling and Testing

(Copies of specifications, standards, handbooks, drawings, and publications and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

American Society for Testing and Materials (ASTM)

ASTM D56 - Flash Point by Tag Closed Tester, Test Method for
ASTM D822 - Operating Light and Water - Exposure Apparatus (Carbon are Type) for Testing Paint, Varnish, Lacquer, and Related Products, Practices for
ASTM D1014 - Conducting Exterior Exposure Test of Paints on Steel, Method of
ASTM D1308 - Effect of Household Chemicals on Clear and Pigmented Organic Finishes, Test Method for
ASTM D1542 - Qualitative Test for Rosin in Varnishes Method of
ASTM D1544 - Color of Transparent Liquids (Gardner Color Scale), Test Method of
ASTM D1545 - Viscosity of Transparent Liquids by Bubble Time Method, Test Method for
ASTM D1639 - Acid Value of Organic Coating Materials, Test Method for
ASTM D1951 - Ash in Drying Oils and Fatty Acids, Test Method for
ASTM D1959 - Iodine Value of Drying Oils and Fatty Acids, Test Method for
ASTM D1962 - Saponification Value of Drying Oils, Fatty Acids, and Polymerized Fatty Acids, Test Method for
ASTM D3272 - Vacuum distillation of Solvents from Solvent-base Paints for Analysis, Method for
ASTM G23 - Operating Light - Exposure Apparatus (Carbon Arc Type) With and Without Water for Exposure of Nonmetallic Materials, Practice for

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St, Philadelphia, PA 19103.)

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(Non-government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supercede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The rust arresting coating furnished under this specification shall be a product which is qualified for listing on the applicable Qualified Products List at the time set for the opening of bids (see 6.4). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under the contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 Composition. The nonvolatile vehicle shall be a mixture of drying and semi-drying oils with driers and shall conform to the requirements specified in Table I.

TABLE I. Composition of nonvolatile vehicle.

	Requirements	
	Minimum	Maximum
Acid number	--	8
Saponification number	185	--
Iodine number	95	115
Rosin	--	Neg
Phenol	--	Neg

3.2.1 Solvents. The volatile portion of the coating shall consist of any suitable petroleum hydrocarbon solvent or solvents. It shall contain no chlorinated hydrocarbons or other solvents of a highly toxic nature.

3.2.1.1 Composition L. The volatile content of composition L coatings shall also conform to the following requirements by volume when tested as in 4.4.3.

(a) Aromatic compounds with eight or more carbon atoms except ethyl benzene: 8 percent maximum.
(b) Ethyl benzene, toluene and branched ketones: 20 percent maximum.
(c) Solvents with an olefinic or cyclo-olefinic type of unsaturation: negative test.
(d) Total of a + b: 20 percent maximum.

3.3 Quantitative requirements. The finished coating material shall conform to the requirements specified in table II.

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TABLE II. Quantitative requirements.

	Requirements	
	Minimum	Maximum
Flash point	38°C. (100°F.)	--
Total solids, percent by weight of compound	58	--
Ash, percent by weight of compound	--	1.0
Viscosity (Gardner Tubes)	E	G
Color (Gardner Color Standards 1953)	--	18
Drying time		
Set to touch, hours	--	3
Dry hard, hours	--	24

3.4 Qualitative requirement.

3.4.1 Condition in container. A freshly opened full container of the coating, tested as in 4.4.5, shall be free from seeds, lumps, abnormal thickening, or livering.

3.4.2 Storage stability. A three quarter filled, closed 8-ounce jar of the coating shall show no skinning when tested as in 4.4.6. After aging as in 4.4.6, the coating shall show no gel formation and the viscosity shall increase no more than one letter on the Gardner viscosity scale.

3.4.3 Dilution stability. When thinned as in 4.4.7, the coating shall remain stable and uniform showing no gel formation.

3.4.4 Knife test. A film of the coating, tested as in 4.4.8, shall ribbon or curl from the metal on cutting and the cut shall show beveled edges.

3.4.5 Brushing properties. The coating, tested as in 4.4.9, shall brush satisfactorily in all respects and shall dry to a smooth glossy uniform film.

3.4.6 Spraying properties. The coating, tested as in 4.4.10, shall spray satisfactorily in all respects and shall present a clear smooth glossy finish.

3.4.7 Boiling water resistance. A film of the coating, tested as in 4.4.11, shall show no wrinkling or blistering and no more than moderate whitening immediately after removal of the panel from the water. When examined 2 hours after removal, the film shall show no whitening and no more than slight dulling.

3.4.8 Cold water resistance. A film of the coating, tested as in 4.4.12, shall show no wrinkling or blistering and no more than slight whitening immediately after removal of the panel from the water. When examined 2 hours after removal, the film shall show no whitening and no more than a slight dulling.

3.4.9 Accelerated weathering. A film of the coating, tested as in 4.4.13, shall show no cracking, checking, peeling, or whitening and no more than a slight loss of gloss.

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3.4.10 Weather resistance. A film of the coating, tested as in 4.4.14, shall show no loss of rust arresting properties. No flaking, peeling, nor renewed rusting shall be evident on completion of the exposure cycle.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Sampling, inspection and testing. Unless otherwise specified, sampling, inspection and testing shall be in accordance with method 1031 of FED-STD-141.

4.3 Classification of tests. Testing under this specification shall be for the purpose of:

(a) Qualification
(b) Acceptance of individual lots.

4.3.1 Qualification testing shall consist of tests for all requirements specified in section 3 (see 6.4).

4.3.2 Acceptance tests shall normally consist of tests for all requirements specified in section 3 with the exception of storage stability (3.4.2) and weather resistance (3.4.10).

4.4 Test methods.

4.4.1 Test conditions. The routine and referee testing conditions shall be in accordance with section 7 of FED-STD-141 except as otherwise specified herein.

4.4.2 The following tests shall be conducted in accordance with FED-STD-141 and as hereinafter specified.

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TABLF III. Index.

Tests	Test Method: Applicable method in Fed. Test Method Std. No. 141	Test Method: Paragraph of this specification giving further references	Paragraph of this specification giving requirements
Acid number	ASTM D1639	--	Table I
Saponification number	ASTM D1962	--	Table I
Iodine number	ASTM D1959	--	Table I
Rosin	ASTM D1542	--	Table I
Phenol	5141	--	Table I
Solvent separation	ASTM D3272	4.4.3.1	3.2.1
Chlorinated solvents	5132.1	--	3.2.1
Aromatic hydrocarbons	7356	4.4.3.2	3.2.1.1
Olefinic and cyclo-olefinic compounds	7356	4.4.3.3	3.2.1.1
Ketones	5172	--	3.2.1.1
Flash point	ASTM D56	--	Table II
Total solids	4042	--	Table II
Ash	ASTM D1951	--	Table II
Viscosity	ASTM D1545	--	Table II
Color	ASTM D1544	4.4.4	Table II
Drying time	4061	--	Table II
Condition in container	3011	4.4.5	3.4.1
Storage stability	3021	4.4.6	3.4.2
Dilution stability	4203	4.4.7	3.4.3
Knife test	6304	4.4.8	3.4.4
Preparation of Steel Panels	2011.2	4.4.8	3.4.4
Brushing properties	4321	4.4.9	3.4.5
Spraying properties	2131	4.4.10	3.4.6
Spraying Properties	4331.1	4.4.10	3.4.6
Boiling water resistance	ASTM D1308 Sec. 5D	4.4.11	3.4.7
Cold water resistance	ASTM D1308 Sec. 5D	4.4.12	3.4.8
Accelerated weathering	ASTM G23, D822	4.4.13	3.4.9
Weather resistance	ASTM D1014	4.4.14	3.4.10

4.4.3 Solvent analysis for composition L.

4.4.3.1 Separation of volatile portion. Separate volatile portion in accordance with ASTM D3272. Reserve collected distillate for the tests for chlorinated solvents, aromatic content, toluene, ethyl benzene, olefinic or cyclo-olefinic compounds, and ketones.

4.4.3.2 Aromatic content. Determine total aromatic content of volatile portion in accordance with procedure A, method 7356 of FED-STD-141. If the total aromatic content is between 8 and 20 percent determine percent of toluene and ethylbenzene in accordance with procedure B, method 7356 of FED-STD-141.

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conditions shall be in accordance with FED-STD-141,

4.4.3.3 Determination of ketone content. Determine total ketone content according to method 5172 of FED-STD-141 (para 4.3); calculate as methyl isobutyl ketone. Omit first five sentences of paragraph 4.3.1 substituting the following. Add 15 ml. of absolute alcohol to a 250 ml. glass stoppered Erlenmeyer flask and accurately pipette in 1.5 ml. of distillate followed by exactly 25 ml. of hydroxylamine hydrochloride reagent.

$$\% \text{ ketone, v/v (computed as methyl isobutyl ketone)} = \frac{(V_1 - V_2) \times N \times 0.10 \times 100}{1.5 \times 0.98 \times 0.82}$$

4.4.4 Color. Reduce 1 part by volume of the packaged coating with 1 part by volume of thinner conforming to TT-T-291, Type 1 Grade A for composition G and Type 11 Grade A for composition L. Test as in ASTM D1544 and observe for compliance with Table II.

4.4.5 Condition in container. Determine package condition as in method 3011 of FED-STD-141 and observe for compliance with 3.4.1.

4.4.6 Storage stability. Allow the coating to stand 30 days and determine the skinning as in method 3021 of FED-STD-141 and observe for compliance with 3.4.2.

4.4.7 Dilution stability. Reduce 3 parts by volume of the coating with 1 part by volume of thinner conforming to TT-T-291, Type 1 Grade A for composition G and Type 11 Grade A for composition L. Then test as in method 4203 of FED-STD-141 for compliance with 3.4.3.

4.4.8 Knife test. Spray clean a 3 by 6 inch steel panel as in procedure A of method 2011.2 of FED-STD-141 by using the aliphatic naphtha ethylene glycol monoethyl ether mixture. Spray a film of the coating on the panel to a dry film thickness between 0.0010 and 0.0012 inches. Air dry the coated panel for 96 hours under referee conditions and perform the knife test as in method 6304 of FED-STD-141. Observe for compliance with 3.4.4.

4.4.9 Brushing properties. Reduce 9 parts by volume of the coating with 1 part by volume of thinner conforming to TT-T-291, Type 1 Grade A for composition G and Type 11 Grade A for composition L. Determine brushing properties as in method 4321 of FED-STD-141 and observe for compliance with 3.4.5.

4.4.10 Spraying properties. Reduce 4 parts by volume of the coating with 1 part by volume of thinner conforming to TT-T-291, Type 1 Grade A for composition G and Type 11 Grade A for composition L. Spray on a steel panel to a dry film thickness between 0.0010 and 0.0012 inches and observe for spraying properties as in method 4331.1 of FED-STD-141 for compliance with 3.4.6. For referee test use automatic application per method 2131 of FED-STD-141.

4.4.11 Boiling water resistance. Prepare a 3 by 6 inch steel panel as in 4.4.8. Air dry 96 hours under referee conditions and then immerse for 15 minutes in boiling water as in ASTM D1308, Sec. 5D. At the end of the test period remove the panel and check for compliance with 3.4.7.

4.4.12 Cold water resistance. Prepare a panel as in 4.4.8. Coat all exposed unpainted metal surfaces with wax or a suitable protective coating.

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Then immerse in distilled water at 23° ± 1°C (73 ± 1°F) for 18 hours as in ASTM D1308, Sec 5D. At the end of the test period remove and examine for compliance with 3.4.8.

4.4.13 Accelerated weathering. Prepare 2 test panels of the coating as in 4.4.8 and expose for 300 hours to accelerated weathering as in ASTM G23 or D822 using a twin arc apparatus. Examine for compliance with the requirements of 3.4.9.

4.4.14 Weather resistance. Solvent clean three 4 by 12 inch steel panels using the brush cleaning procedure of method 2011.2 of FED-STD-141. Place them on outdoor exposure for at least 3 months at an angle of 45° facing south in the latitude of Washington, D. C. When a heavy uniform coating of rust has formed, wire brush the panels to remove excess and loose rust before application of the rust arresting coating. Brush on a film of the rust arresting coating in excess to give a continuous film with no dry spots. Stand the panel on one end in a vertical position and allow the excess coating to drain off. Allow to air dry 48 hours under referee conditions and place on outdoor exposure for 9 months at an angle of 45° facing south in the vicinity of Washington, D. C. as in ASTM D1014. At the end of the exposure period inspect the panels for compliance with 3.4.10.

4.5 Inspection of the preparation for delivery. The packaging, packing and marking of the rust arresting coating shall be inspected to determine compliance with the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 The rust arresting coating shall be delivered in 1-quart or 1-gallon rectangular containers, in 5-gallon tight head pails or in 55-gallon steel drums as specified (see 6.2). The coating shall be packaged in level A, B or C; packed level A, B, or C as specified (see 6.2) in accordance with PPP-P-1892.

5.2 Marking. The containers shall be marked in accordance with PPP-P-1892. In addition, each container shall be marked with the manufacturer's Qualified Products List designation.

6. NOTES

6.1 Intended use. The coating covered by this specification is intended primarily for touching up rusted areas on various types of materials and equipment during outdoor storage. It is intended for temporary protection until proper refinishing can be accomplished.

6.2 Ordering data. Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Composition required (see 1.2).
(c) Whether inspection will be made in accordance with section 1031 of FED-STD-141 (see 4.2).
(d) Size of pails or cans (see 5.1).
(e) Levels of packaging and packing required.
(f) The coating should be purchased by volume, the unit being one U.S. liquid gallon of 231 cubic inches at 68°F. (20°C.).

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6.3 Surface preparation. Prior to application of the rust arresting coating all loose rust, scale, paint, etc. should be removed. Removal of all rust and paint is not expected however, since the coating to be applied is a temporary protective coating.

6.4 Qualification. With respect to products requiring qualification, awards will be made only for such products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the US Army Belvoir Research and Development Center, ATTN: STRBE-TSF, Fort Belvoir, VA 22060-5606.

6.5 Classification. Composition L coatings should be specified for use in areas with regulation controlling the emission of solvents into the atmosphere.

6.6 Subject term (key word) listing.

Coating
Corrosion
Metal
Rust
Weather Resistance

Custodians:
Army - MR
Navy - SH
Air Force - 84

Review activities:
Navy - SH

User activities:
Army - AT
Navy - OS, MC

Preparing activity:
Army - MR

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-10036E	R UST ARRESTING COATING (FOR TREATMENT OF RUSTED METAL)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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